

SinhalaJournal-LLM

A Style-Adaptive Large Language ✶ Model Framework for Sinhala Journalism

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering
Project ID: R26-SE-037

Member 1 – Sinhala Grammar Checker
Member 2 – Sinhala Headline & Image Generator
Member 3 – Sinhala News Summarizer
Member 4 – Sinhala Article Style Rewriter

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SinhalaJournal-LLM - A Style-Adaptive Large Language Model Framework for Sinhala Journalism

Problem

Lack of accurate and intelligent **AI-powered** tools for **summarizing, correcting, rewriting, and generating** Sinhala news content due to Sinhala being a **low-resource language** with **limited NLP support**.

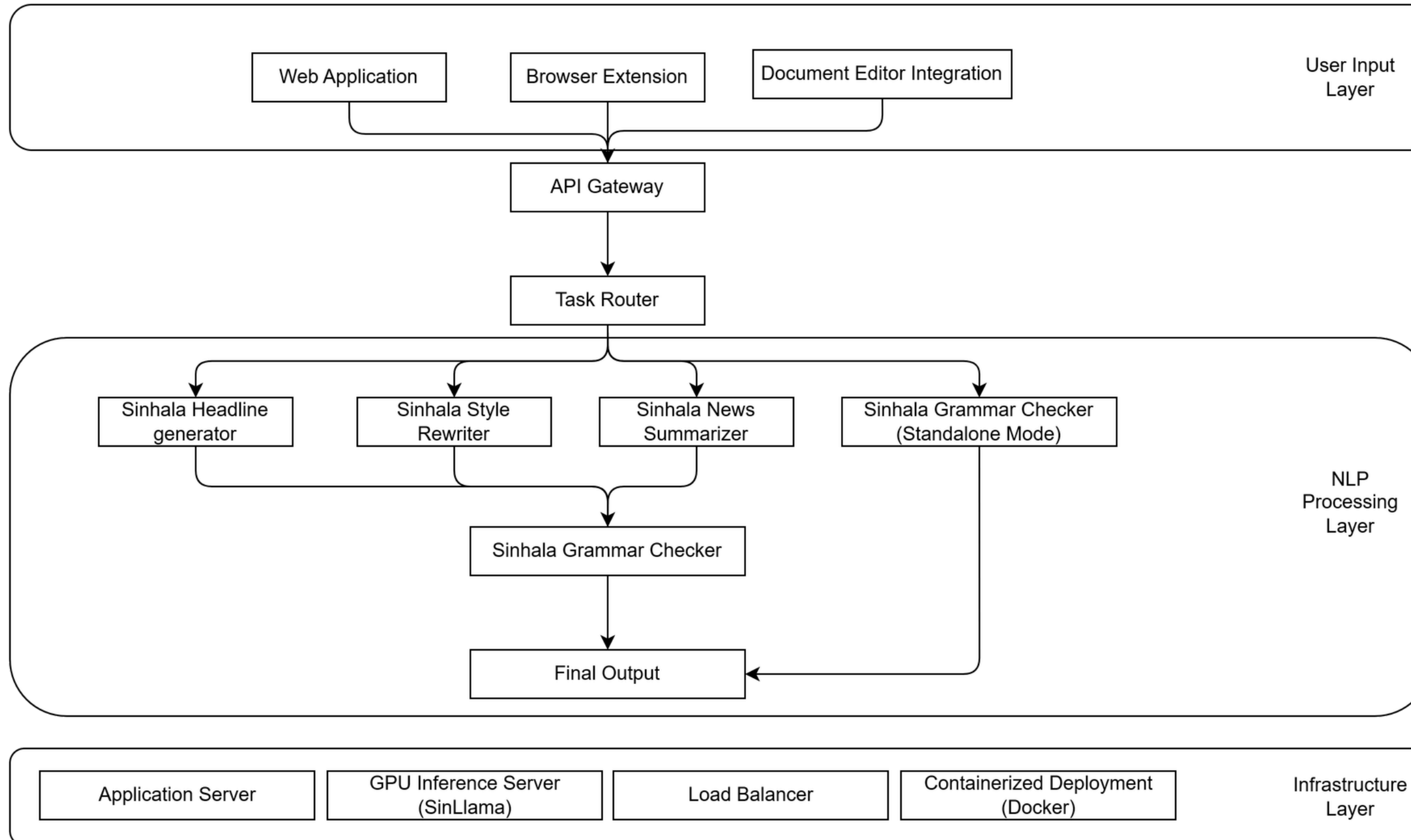
Novelty

First unified **Sinhala journalism-focused LLM** system that combines grammar correction, headline generation, style-aware rewriting, and news summarization using **fine-tuned Sinhala** low-resource language models and **domain-specific news datasets**

Users

- Journalists and media organizations
- Students and researchers
- Sinhala news readers
- Digital news platforms and publishers

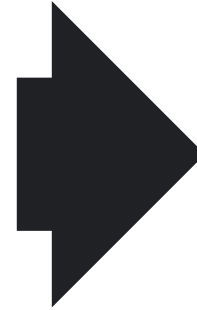
High-Level Architecture



ADDRESSED PANAL COMMENTS

OVERALL PANAL COMMENT

Sinllama can be **directly** used for Grammar checking, headline generation, summarization and style rewrite



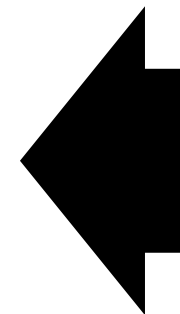
OUR EVALUATION

- We run sinllama on the server and we prompted a simple greeting “කොහොමද” it produced random words, emojis, Siri mentions, and unrelated continuation.
- We prompted simple tasks for each of our component, it showed very Poor Instruction Following, Hallucination Problems, Journalism Quality Failure.



OUR SOLUTION

- We fine tuned 4 components and we had a positive progress on performing the same tasks better than sinllama model



OUR OBSERVATION

- Sinllama model is hallucinating
- Random outputs
- Not following instructions
- Garbage values

Why use Sinllama?

Model	Type	Can Generate Text?	Sinhala-Specific?
SinBERT	Encoder (RoBERTa)	No	✓ Yes
XLM-R	Multilingual Encoder	No	0.15% Sinhala data only
mT5	Seq2Seq Multilingual	Weak	Minimal
LLaMA 3 (raw)	Decoder LLM	✓ Yes	Incoherent Sinhala output
SinLlama	Decoder LLM + DAPT	✓ Yes	✓ Full Sinhala adaptation

SinLlama is the only open-source Sinhala generative LLM with **Sinhala pretraining**, **custom tokenization**, and **LoRA support** making it the best base model for SinhalaJournal-LLM.



Sinhala Grammar Checker & corrector

Specialization - Software Engineering



Sinhala Grammar Checker & corrector

Sub Problem

Existing Sinhala tools mainly support **only basic spell checking** and **cannot handle deeper grammar** correction capabilities for **newsroom-style** content

Novelty

Journalism-focused Sinhala grammar correction using **Sinhala large language models** and context-aware NLP techniques.



Sinhala Grammar Checker & corrector

How the Novelty solved the Sub-Problem

- Detect **grammar**, **spelling**, and **sentence errors** in Sinhala news articles.
- Suggests corrected sentences

How the Solution Helps Users

- Helps users **write** and **recorrect** professional Sinhala news articles more easily.
- provides **fast grammar correction** , **reduces editing time**, and improves writing consistency.
- Provide **web-based interface** , **chrome** and **google doc plugin** for article correction



Sinhala Grammar Checker & corrector

System overview

Input Article



Sentence splitting



Text Preprocessing



LLM-based Grammar Correction
(SinLLaMA + grammar LoRA)



Sentence rejoining



Final Corrected Output

මම ඊයේ පාසලට යනවා. එහිදී මම ගොඩක් දේවල් ඉගෙනගමු

[මම ඊයේ පාසලට යනවා.][එහිදී මම ගොඩක් දේවල් ඉගෙනගමු]

[මම][ඊයේ][පාසලට][යනවා.] [එහිදී][මම][ගොඩක්][දේවල්][ඉගෙනගමු]

SinLLaMA + grammar LoRA - fix each sentence

මම ඊයේ පාසලට ගියෙමි.+එහිදී මම ගොඩක් දේවල් ඉගෙනගත්තෙමි

මම ඊයේ පාසලට ගියෙමි. එහිදී මම ගොඩක් දේවල් ඉගෙනගත්තෙමි



Style-Adaptive Headline & Image Generator

Specialization - Software Engineering

Style-Adaptive Headline & Image Generator

Sub Problem

Sinhala digital news platforms rely on **manually** written **headlines and images**, causing **inconsistency** and **slower** publishing

Novelty

Style-adaptive **Sinhala headline generation** using a fine-tuned SinLlama LLM and aligned **image generation** via multimodal AI APIs.

Style-Adaptive Headline & Image Generator

How the Novelty solved the Sub-Problem

- Automatically generates style-controlled **headlines** from Sinhala news articles
- Supports **formal, breaking news, and youth writing styles** etc.

How the Solution Helps Users

- Journalists need **fast, accurate, and context-aware** headlines in Sinhala Unicode
- Delivers results instantly through an easy-to-use web-based interface
- No **manual headline writing** effort required from the journalist
- Automatically generates a **thumbnail image**, saving time on visual content

Style-Adaptive Headline & Image Generator



කොළඹ නගරයේ නව බස් මාර්ගයක්
හඳුන්වා දීම

["කොළඹ", "නගරය", "නව", "බස්", "මාර්ගය"]

Style = Breaking News / Formal / Youth Style

කොළඹ නව බස් මාර්ගයක් හඳුන්වාදෙයි

කොළඹ නව බස් සේවාවක් - මාර්ගයක් අද
හඳුන්වාදෙයි

Entities → "කොළඹ", "බස් සේවාව"
Keywords → "නව මාර්ගය", "ප්‍රවාහන"

City bus traveling in Colombo urban road,
public transport news scene



Sinhala News Summarizer

Specialization - Software Engineering



Sinhala News Summarizer

Sub-Problem

Existing Sinhala summarizers **just copy sentences** from the article. The summaries break context and don't flow well. **AI models** also sometimes **generate wrong facts**.

Novelty

Our system writes summaries in its own words, **keeps the original meaning**, and sounds like real Sinhala news writing using **Abstractive Generation**.



Sinhala News Summarizer

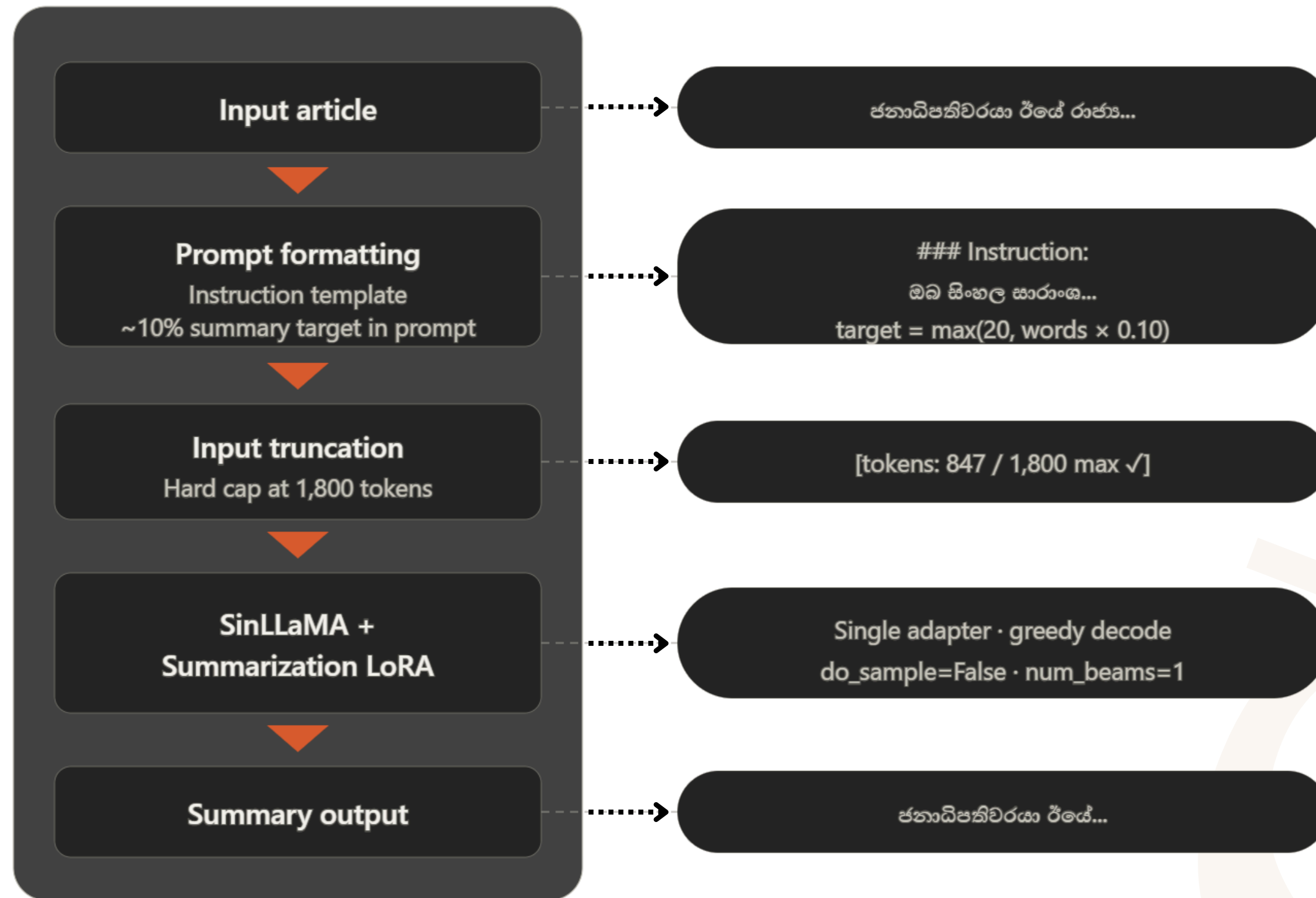
How the Novelty solves the Sub-Problem

It takes a long Sinhala news article and produces a short, clear summary **using SinLlama + LoRA**. Makes articles easier and faster to read.

How this Solution Help Users

- Accurate summaries
- Saves reading time
- Handles Sinhala text properly
- Simple web interface
- **Keeps the meaning**

Sinhala News Summarizer





Sinhala Article Style Rewriter

Specialization - Software Engineering

Sinhala Article Style Rewriter



Sub-Problem

- Existing Sinhala NLP tools do not support article rewriting and style adaptation for journalism content.
- Media organizations manually rewrite articles for different audiences and platforms,

Novelty

- Style-controlled Sinhala news article rewriting using AI
- Uses a **LoRA fine-tuned** Sinhala Large Language Model
- Transforms the same article into multiple journalism writing styles
- Preserves semantic meaning and factual accuracy during rewriting

Sinhala Article Style Rewriter

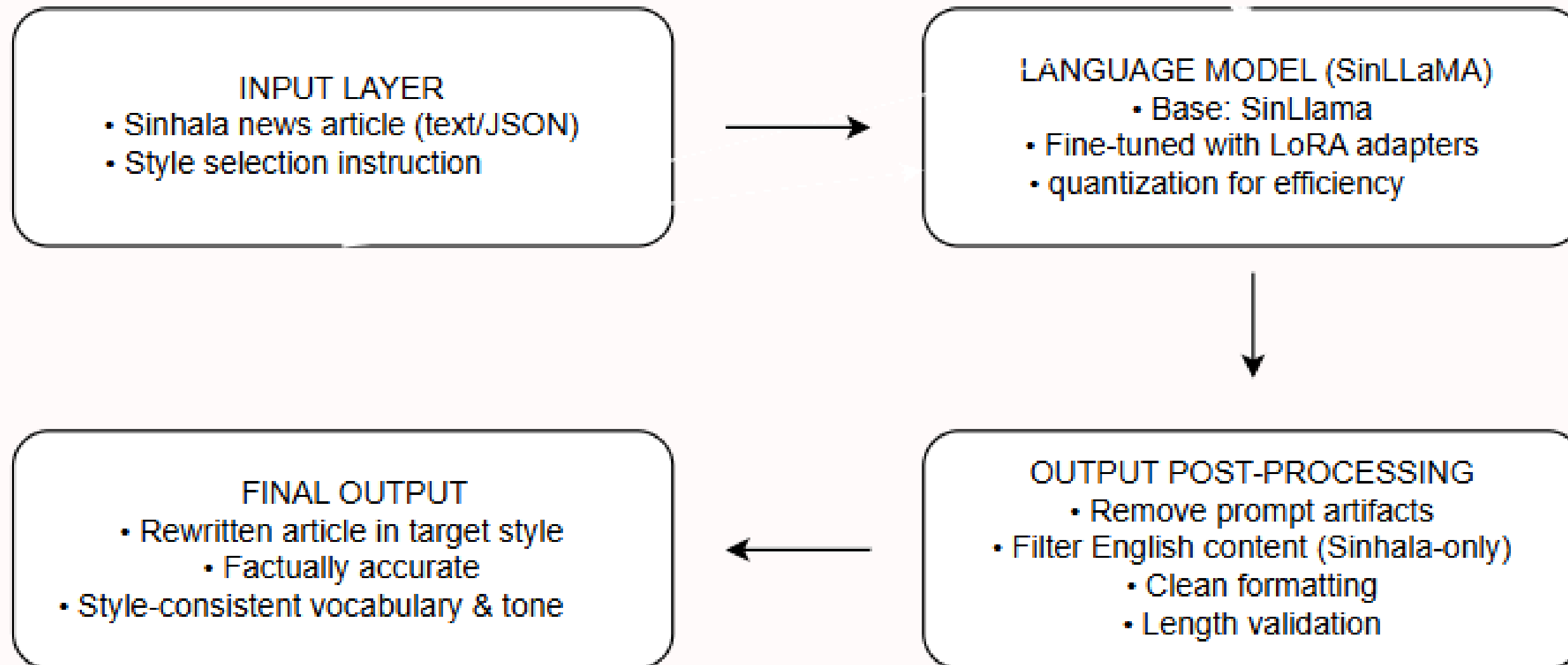
How the Novelty solved the Sub-Problem

- Rewrites Sinhala news articles into different journalism styles.
- Reduces manual effort.
- Preserves the original meaning while changing the writing tone.
- Helps media organizations quickly generate content.

How the Solution Helps Users

- Helps users rewrite Sinhala news articles into different journalism styles
- Preserves the original meaning and factual accuracy of articles
- Reduces manual effort and supports fast content adaptation
- Provides a simple and user-friendly web interface for rewriting content

Sinhala Article Style Rewriter



Commercialization Plan

TARGET MARKET

- Sinhala journalists
- Media organizations
- Journalism Schools / Media Training Institutes

REVENUE MODEL

- SaaS Subscription
- Plus plan for individual journalists
- Pro plan for media organizations

VALUE PROPOSITION

- Write summaries faster
- Reduce manual editing work
- Maintain consistent news style
- Publish news quicker

What is Unique - First unified Sinhala journalism-focused LLM platform

Combines:

- Sinhala Grammar Checker
- Sinhala Headline Generator
- Sinhala Style Rewriter
- Sinhala News Summarizer

Uses:

- Sinhala-specific datasets
- Fine-tuned Sinhala LLM models
- Journalism-domain adaptation

Multiple newsroom writing styles:

- Formal
- Editorial
- Sports
- Youth/news portal style

Thank
You